

From Raw Sanger Chromatograms to Phylogenetic Analysis: A Beginner Bioinformatics Workflow Using Fungal ITS Sequences

Project Overview

This project demonstrates an end-to-end beginner bioinformatics workflow using raw Sanger sequencing chromatogram data obtained from a publicly available Zenodo dataset. The workflow includes chromatogram quality assessment, sequence trimming, sequence alignment, BLAST identification, and phylogenetic analysis using fungal ITS sequences.

Objective

The objective of this project was to understand how raw sequencing data are processed into biologically interpretable results using standard bioinformatics tools and workflows.

Workflow Summary

- Selection of raw .ab1 chromatogram files from Zenodo dataset
- Chromatogram quality assessment in Chromas
- Forward and reverse read alignment in MEGA12 using MUSCLE
- Manual trimming of low-quality and non-overlapping regions
- Generation of consensus sequence
- NCBI BLAST analysis
- Retrieval of top matching ITS reference sequences
- Multiple sequence alignment
- Neighbor-Joining phylogenetic tree construction with bootstrap analysis

Chromatogram Quality Assessment Screenshot

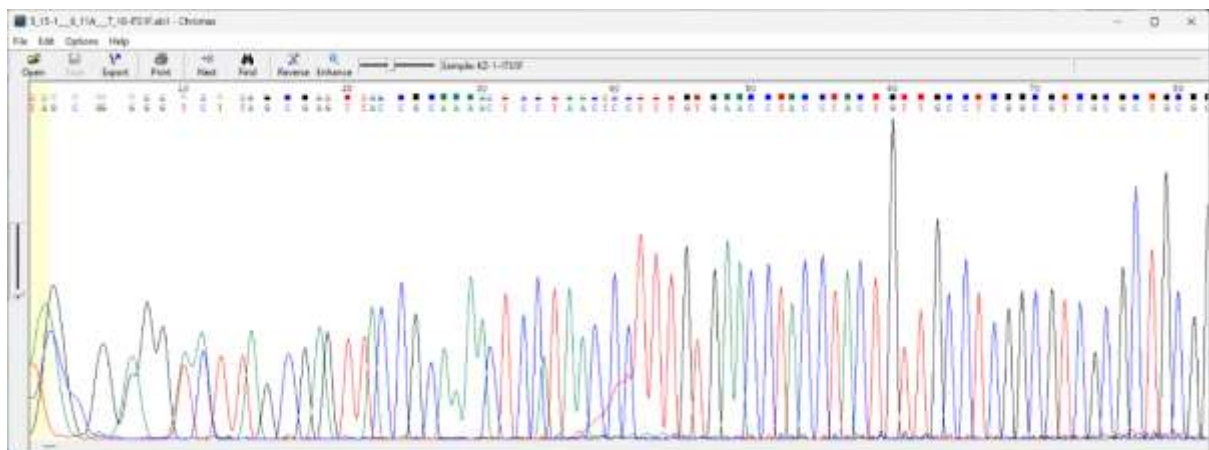


Figure 1: Chromatogram showing poor quality read in the beginning of a forward sequence

The top screenshot displays the M12 Alignment Explorer interface with a single alignment. The DNA sequence is labeled 'Spermidinase' and the translated protein sequence is shown below it. The alignment is visualized with a color-coded bar representing the protein sequence.

The bottom screenshot shows the same interface with a multiple sequence alignment. The DNA sequence is labeled 'Spermidinase' and the translated protein sequence is shown below it. The alignment is visualized with a color-coded bar representing the protein sequence. The sequences are aligned side-by-side, showing the conservation of amino acids across different species.

BLAST » blastn suite » results for RID-2C64SSYC014

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< Edit Search Save Search Search Summary ▼

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Filter Results

Organism only top 20 will appear ☐ exclude

[+ Add organism](#)

Percent Identity to **E value** to **Query Coverage** to

Filter **Reset**

Job Title	Sample 1
RID	2C64SSYC014 Search expires on 06-09 09:19 am Download All ▼
Program	BLASTN Citation ▼
Database	core_rnt See details ▼
Query ID	IcdQuery_1741251
Description	KZ-1-ITS1R
Molecule type	dna
Query Length	202
Other reports	Distance tree of results MSA viewer ?

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments

Download Select columns Show 100 ?

☐ select all 77 sequences selected

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Pct Ident	Acc Len	Accession
✓	Annuobryconium sp. isolate CSC21B0584 small subunit ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	619	FJ481155.1
✓	Annuobryconium annuum isolate N32 18S ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	640	FJ481155.1
✓	Annuobryconium annuum isolate N39 18S ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	604	FJ481155.1
✓	Annuobryconium sp. CF_K220.1 genes for 18S rRNA, ITS1, 5.8S rRNA, partial and complete sequence	Annuobryconium	374	374	100%	4e-99	100.00%	279	LC793340.1
✓	Annuobryconium annuum genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete locus	Annuobryconium	374	374	100%	4e-99	100.00%	821	LC317411.1
✓	Annuobryconium sp. isolate JQDH192-2 internal transcribed spacer 1, partial sequence, 5.8S ribosomal RNA gene, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	589	CN507326.1
✓	Annuobryconium sp. HMF382 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence	Annuobryconium	374	374	100%	4e-99	100.00%	617	LC428833.1
✓	Annuobryconium annuum voucher 14143 small subunit ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	811	MY023967.1
✓	Funari sp. isolate ARIZ-Teneva BE5041 internal transcribed spacer 1, partial sequence, 5.8S ribosomal RNA gene, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Funari sp.	374	374	100%	4e-99	100.00%	1071	FQ576784.1
✓	Annuobryconium annuum strain MQ313254 small subunit ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	658	MH558286.1
✓	Funari endochytrid strain SV1857 18S ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Funari endochytrid	374	374	100%	4e-99	100.00%	1176	XV795279.1
✓	Annuobryconium annuum isolate AH258 small subunit ribosomal RNA gene, partial sequence, internal transcribed spacer 1, partial sequence, internal transcribed spacer 2, complete sequence, and 5.8S rRNA	Annuobryconium	374	374	100%	4e-99	100.00%	627	FP357844.1

Phylogenetic Tree Screenshot

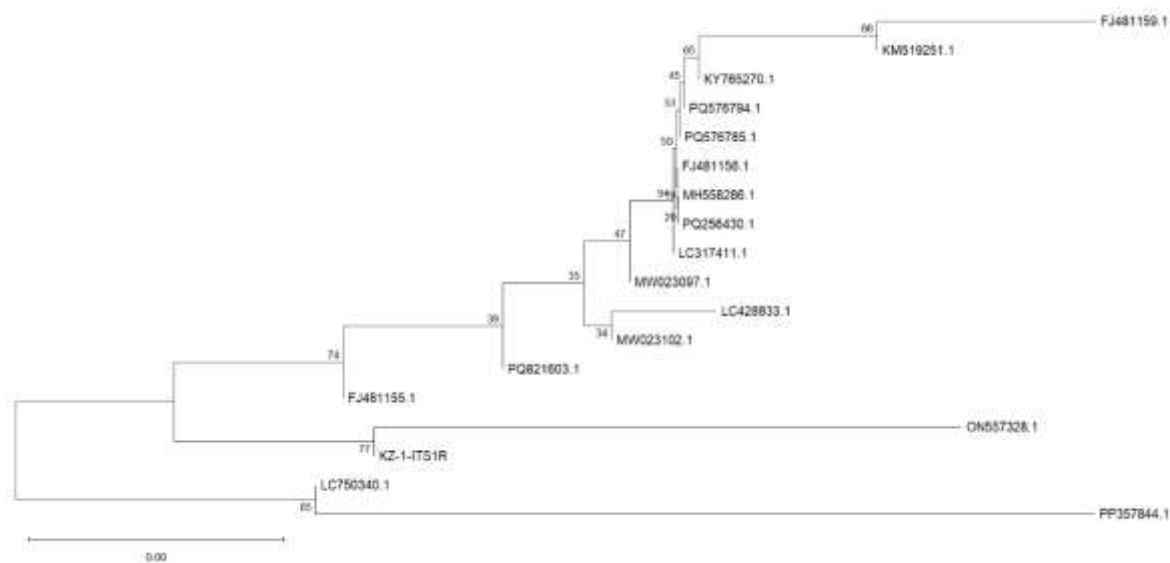


Figure 3: Phylogenetic Tree produced using MEGA12 showing the relationship of sample with other species

Results and Interpretation

BLAST analysis showed strong similarity to *Annulohypoxylon*-associated fungal ITS sequences with 100% query coverage and identity among top hits. Phylogenetic analysis placed the sample within an *Annulohypoxylon*-related clade and demonstrated evolutionary relatedness with selected reference sequences.

Tools Used

Tools	Purpose
Chromas	Chromatogram visualization and quality assessment
MEGA12	Sequence alignment and phylogenetic analysis
MUSCLE	Multiple sequence alignment
NCBI BLAST	Sequence similarity search and identification

Conclusion

This project successfully demonstrated a complete beginner-level Sanger sequencing bioinformatics workflow, beginning with raw chromatogram data and ending with phylogenetic interpretation. The project provided practical experience in sequence quality control, sequence alignment, BLAST analysis, and evolutionary analysis using phylogenetic trees.